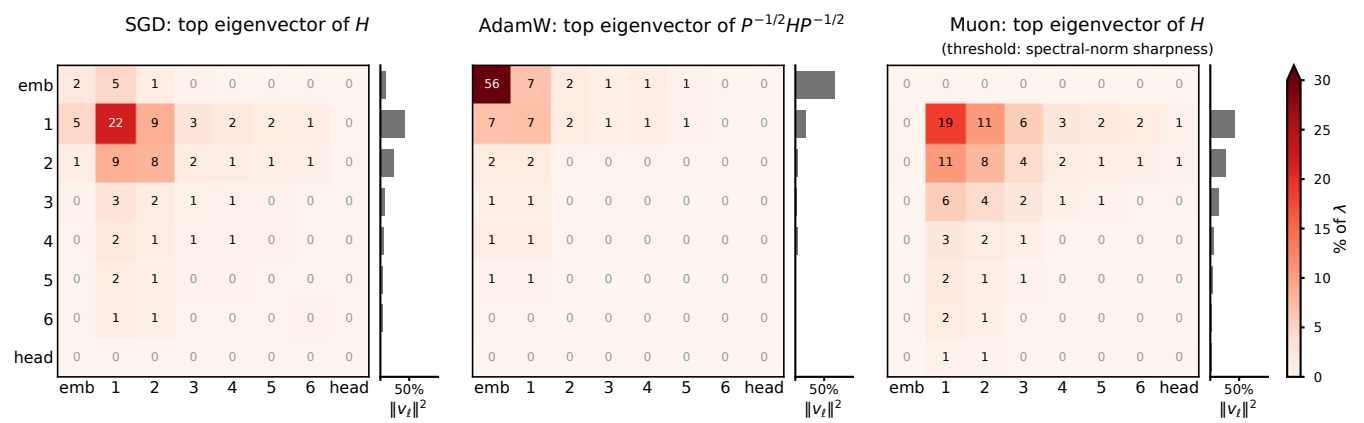
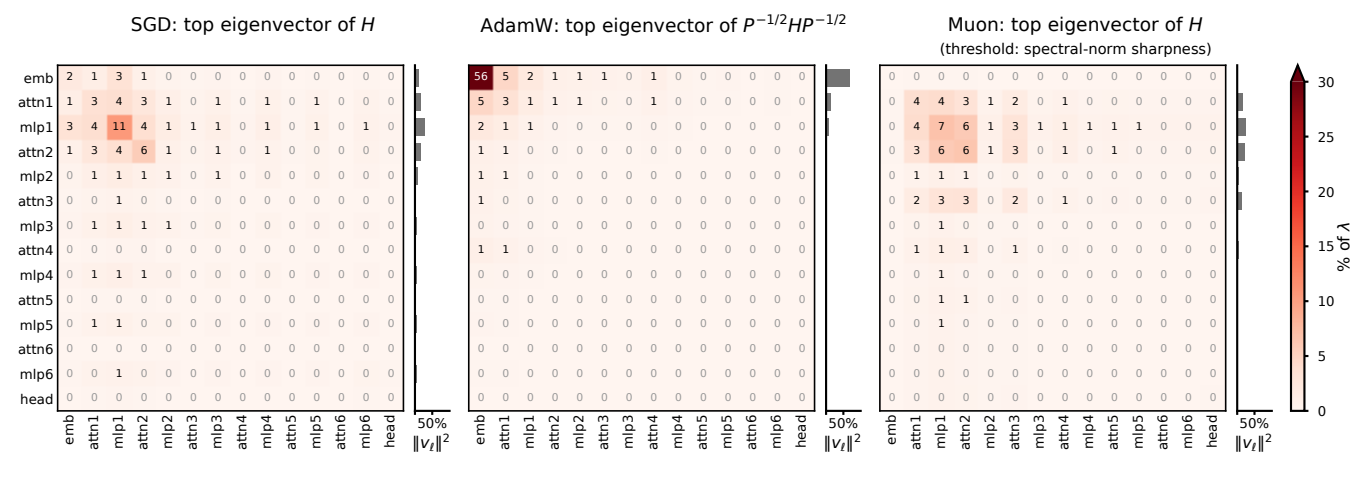

SGD: top eigenvector of H
AdamW: top eigenvector of $P^{-1/2}HP^{-1/2}$
Muon: top eigenvector of H
(threshold: spectral-norm sharpness)
emb 1 2 3 4 5 6 head
50% $\|v_\ell\|^2$
% of λ
0 5 10 15 20 25 30


SGD: top eigenvector of H
AdamW: top eigenvector of $P^{-1/2}HP^{-1/2}$
Muon: top eigenvector of H
(threshold: spectral-norm sharpness)
emb attn1 mlp1 attn2 mlp2 attn3 mlp3 attn4 mlp4 attn5 mlp5 attn6 mlp6 head
50% $\|v_\ell\|^2$
% of λ
0 5 10 15 20 25 30